What are the most effective methods and tools that organizations use to achieve compliance with the EU AI Act?

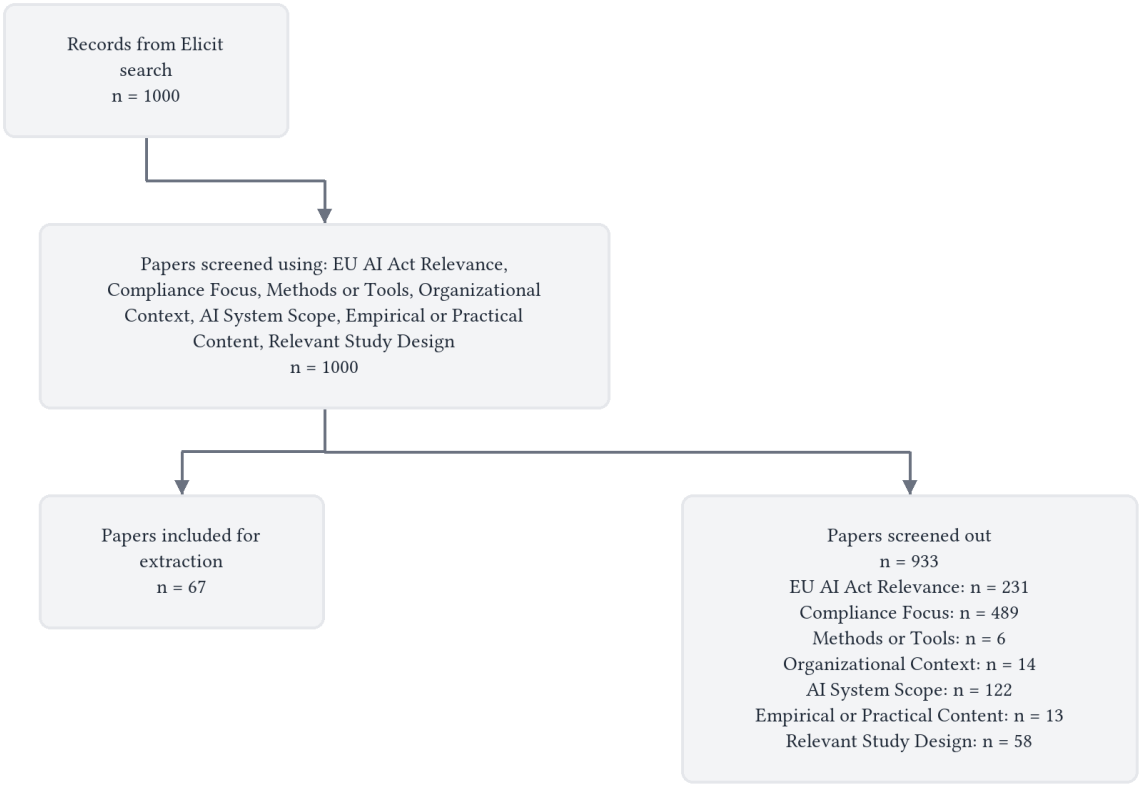
No single method has been field-validated as comprehensively effective, but the strongest evidence supports a layered approach combining compliance-by-design controls embedded during model development (particularly adversarial debiasing for fairness and SHAP-based explainability with governance approval gates), LLM-assisted documentation tools with mandatory human legal review, and structured institutional governance frameworks that assign clear oversight roles, scenario-based staff training, and continuous post-deployment monitoring—recognising that technical tools alone are insufficient without the organisational capacity to enforce them.

Abstract

Evidence from 67 sources spanning framework proposals, empirical evaluations, and case studies identifies six broad categories of compliance approaches: AI-assisted documentation tools, technical documentation frameworks, fairness and explainability methods, organisational governance structures, post-market monitoring systems, and standards-based compliance pathways. No single method or tool has been demonstrated to achieve full EU AI Act compliance across the lifecycle in a field-validated, real-world setting. LLM-based documentation tools such as DoXpert and GPT-4-based checkers can identify gaps in technical documentation, with DoXpert achieving 81.54–83.08% accuracy in iterative draft review and 84.6–96.9% agreement with expert judgments [1], but they perform poorly on nuanced legal reasoning and technical conformity verification and require human expert oversight [1, 2]. Among bias mitigation methods, adversarial debiasing dominates re-weighing and reject-option classification in insurance underwriting, closing 65–82% of fairness gaps while adding only 14 basis points to solvency capital requirements [3], though this finding is domain-specific and the mitigation requires continuous monitoring [3]. Structured organisational compliance frameworks—combining risk-classification guidance, scenario-based training, obligation checklists, and assigned governance roles—were associated in one manufacturing pilot with increases in risk-identification accuracy from 40% to 86% and regulatory knowledge from 42% to 81% ($p<0.01$) [4], though the absence of a control group limits causal attribution. A critical methodological hazard for fairness auditing is that using biased evaluation labels can reverse the apparent effectiveness of interventions such as SMOTE by as much as 11 percentage points [5], meaning currently accepted fairness metrics may not reliably constitute compliance evidence without explicit evaluation-bias detection [5].

The evidence consistently supports three cross-cutting conclusions. First, compliance-by-design—embedding fairness, explainability, logging, and governance controls during development rather than retrofitting them post-deployment—is more technically tractable and less costly than after-the-fact remediation [6–8]. Second, technical tools alone are insufficient: effective compliance requires institutional governance capacity, designated oversight roles, and the practical ability to understand, justify, correct, and suspend deployed systems [9–11]. Third, a persistent documentation-implementation gap means that high governance-document scores do not reliably reflect actual safeguard implementation, with technical robustness and safety consistently the lowest-scoring governance dimension in documented institutional assessments [12–14]. SMEs and low-capacity actors face structurally greater compliance burdens [13, 15, 16], and the most significant capability gap across the literature is the absence of field-validated, sector-specific compliance tools that integrate the full AI Act lifecycle within the resource constraints of these actors [13].

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What are the most effective methods and tools that organizations use to achieve compliance with the EU AI Act?"

We applied the following filters: journal quartile Q3 or better; with PDF available.

The search returned 1,000 total results from Elicit.

We retrieved 1000 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **EU AI Act Relevance**: Does the paper explicitly address the EU AI Act (or its regulatory framework, requirements, or obligations)?
- **Compliance Focus**: Does the paper focus on achieving, supporting, or assessing compliance with AI regulation (rather than merely describing or critiquing the regulation in abstract)?
- **Methods or Tools**: Does the paper discuss specific methods, tools, frameworks, techniques, or approaches used to achieve regulatory compliance?
- **Organizational Context**: Does the paper address compliance efforts at the level of organizations, companies, or institutions (e.g., developers, deployers, or operators of AI systems)?

- **AI System Scope**: Does the paper concern AI systems or AI-enabled products/services subject to regulatory requirements (rather than non-AI software or unrelated technologies)?
- **Empirical or Practical Content**: Does the paper include empirical findings, case studies, practical guidance, or tool evaluations (rather than being purely theoretical or philosophical)?
- **Relevant Study Design**: Is the paper a primary study, systematic review, meta-analysis, or structured literature review (rather than an editorial, opinion piece, or non-peer-reviewed commentary)?

Papers that failed any criterion were automatically excluded; papers with a Maybe on any criterion were screened in only when their relevance rating was at least 4 out of 5.

At abstract screening, the number of papers excluded for each primary reason was:

- **EU AI Act Relevance**: n = 231
- **Compliance Focus**: n = 489
- **Methods or Tools**: n = 6
- **Organizational Context**: n = 14
- **AI System Scope**: n = 122
- **Empirical or Practical Content**: n = 13
- **Relevant Study Design**: n = 58

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Compliance Method**:

  Extract a comprehensive description of the specific method, tool, or framework proposed or evaluated for achieving EU AI Act compliance, including:

- Name of the method/tool/framework (if applicable)
- Core technical or procedural approach (e.g., LLM-based auditing, ontology/knowledge graph, modular governance framework, documentation assistance tool, standardisation process)
- Level of automation: fully automated, semi-automated, or manual/human-led
- Key components or sub-modules and how they interact
- Whether it is compliance-by-design (built into development) or compliance-after-the-fact (auditing/verification)
- Whether it is a novel contribution or an evaluation/application of existing tools

- **EU AI Act Coverage**:

  Extract which specific EU AI Act obligations or provisions the method/tool is designed to address, including:

- Specific articles, chapters, or requirements of the EU AI Act targeted (e.g., risk classification, technical documentation, transparency, human oversight, data governance, post-market monitoring, conformity assessments)
- AI risk tier(s) addressed (unacceptable, high-risk, limited-risk, minimal-risk, GPAI)
- Whether it covers the full compliance lifecycle or specific phases (design, deployment, post-deployment)
- Any other regulatory frameworks integrated alongside the EU AI Act (e.g., GDPR, harmonised technical standards such as ISO/IEC 42001, sector-specific regulations, NATO/defence frameworks)

- **Effectiveness Evidence**:

  Extract all evidence provided for the effectiveness of the method/tool in achieving EU AI Act compliance, including:

- How effectiveness was defined and measured (e.g., coverage of regulatory requirements, accuracy of gap identification, expert agreement, audit scores, performance benchmarks)
- Validation approach used (e.g., case studies, expert evaluation, LLM-as-a-Judge, statistical analysis, interviews, proof-of-concept)

- Key quantitative or qualitative results reported (e.g., faithfulness scores, correlation coefficients, coverage rates, cost estimates, time savings)
- Comparison to baseline or alternative approaches, if any
- Limitations or caveats noted by the authors regarding the validity or generalisability of the effectiveness evidence

- **Applicability Context**:

  Extract information about the organisational and operational context in which the method/tool is intended to be used or was tested, including:

- Industry sector(s) targeted or tested (e.g., telecommunications, cybersecurity, healthcare, finance, defence, legal tech, manufacturing)
- Organisation type and size (e.g., SMEs, start-ups, large enterprises, public bodies)
- Type of AI system the method applies to (e.g., high-risk AI, dual-use AI, GPAI models, specific application domains such as intrusion detection, anomaly detection)
- Stage of AI system lifecycle addressed (development, pre-deployment, deployment, post-market monitoring)
- Geographic or jurisdictional scope (EU-focused, global, specific member states)

- **Implementation Challenges**:

  Extract all challenges, barriers, costs, or limitations identified in implementing the method/tool for EU AI Act compliance, including:

- Technical challenges (e.g., scalability, integration with existing pipelines, interpretability of outputs)
- Resource requirements (e.g., cost, time, specialised expertise needed in law and/or technology)
- Organisational barriers (e.g., SME capacity constraints, participation imbalances in standardisation, fragmentation across tools)
- Regulatory or legal ambiguities that hinder implementation (e.g., unclear timelines, overlapping obligations, dual-use complexity)
- Gaps or open problems explicitly acknowledged by the authors

- **Study Design**:

  Extract the research methodology used in the study, including:

- Type of study (e.g., empirical evaluation, proof-of-concept, qualitative interviews, case study, literature/standards review, framework proposal)
- Data or evidence sources used (e.g., number and type of use cases, datasets, number and profile of interview participants, documents analysed)
- Whether the study involves real-world deployment or a controlled/synthetic setting
- Evaluation criteria and who performed the evaluation (e.g., domain experts, automated judges, statistical tests)

- **Key Findings**:

  Extract the authors' main conclusions about what works and what does not for EU AI Act compliance, including:

- Primary takeaways about the effectiveness or utility of the method/tool for EU AI Act compliance
- Recommendations made for organisations seeking compliance (e.g., which tools to adopt, which approaches to combine, policy recommendations)
- Identified best practices or success factors for compliance
- Stated future work or missing capabilities that would need to be developed

Results

Characteristics of Included Studies

The 67 included sources span a wide range of study designs, sector contexts, and compliance approaches. The table below provides a comprehensive overview of each study's design, primary sector, target AI risk tier, and compliance

orientation.

Study	Full text retrieved?	Study type	Primary sector	Risk tier focus	Compliance orientation
Francesco Sovrano et al. (2025) [1]	Yes	Empirical evaluation / proof-of-concept	Finance, healthcare [1]	High-risk [1]	Documentation drafting / after-the-fact review [1]
M. Kauffman et al. (2026) [4]	Yes	Mixed-methods empirical evaluation with pilot	Manufacturing (UK, Brazil) [4]	All risk tiers [4]	Compliance-by-design / organisational readiness [4]
Maria Assunta Cappelli et al. (2026) [2]	Yes	Empirical case-study evaluation	Recruitment, healthcare, urban management, retail [2]	High-risk [2]	Pre-audit self-assessment [2]
R. Kilian et al. (2025) [15]	Yes	Mixed-methods qualitative / standards review	Cross-sector (mobility, finance, healthcare, defence, legal) [15]	High-risk [15]	Harmonised standards / organisational implementation [15]
António Gonçalves & A. Correia (2026) [6]	Yes	Design Science Research / proof-of-concept	Cybersecurity, privacy, critical infrastructure [6]	High-risk [6]	Compliance-by-design (MLOps) [6]
Giedrė Sabaliauskaitė et al. (2026) [17]	No (abstract only)	Framework proposal / case study	Defence and civilian dual-use [17]	Not specified [17]	Compliance-by-design (ontology) [17]
Claudia Maussner et al. (2024) [18]	No (abstract only)	Comparative ML model evaluation	Water supply / critical infrastructure [18]	High-risk [18]	Technical compliance evaluation [18]
B. Jørgensen & Z. Ma (2026) [19]	No (abstract only)	Framework proposal / thematic synthesis	Critical energy systems [19]	Not specified [19]	Compliance-by-design (transparency) [19]
D. Oniga et al. (2026) [20]	Yes	Applied system development / pilot	Border management and security [20]	High-risk (reviewed) [20]	Ethical-by-design / pilot assessment [20]
Siddharth Mahajan et al. (2025) [3]	Yes	Quantitative empirical evaluation	Life and health insurance [3]	High-risk [3]	Development-stage bias mitigation [3]
Hyun-Kyung Lee et al. (2026) [21]	No (abstract only)	Expert-informed framework proposal (AHP + interviews)	General / high-stakes socio-technical [21]	Not specified [21]	Pre-design governance [21]
Imo Enang et al. (2026) [22]	Yes	Qualitative multi-method / framework proposal	Finance, energy, consumer goods, public services [22]	High-risk (Article 14) [22]	Compliance-by-design governance [22]

Study	Full text retrieved?	Study type	Primary sector	Risk tier focus	Compliance orientation
F. Manganello et al. (2025) [12]	Yes	Cross-sectional document analysis / exploratory validation	Higher education (Italy) [12]	High-risk (Annex III, 3b) [12]	After-the-fact governance self-assessment [12]
Vipin Kalra (2025) [23]	Yes	Empirical multi-organisation case study	Contact centres (CCaaS) [23]	High-risk [23]	Lifecycle compliance / XAI governance [23]
Isabelle Hupont et al. (2023) [24]	Yes	Framework proposal / co-design and questionnaire validation	Cross-sector [24]	All risk tiers [24]	Preliminary risk assessment / documentation [24]
Renato Cuocolo et al. (2025) [25]	Yes	Modified Delphi expert-consensus study	Healthcare / radiology [25]	High-risk [25]	Post-market surveillance [25]
Mikel Aramburu et al. (2026) [26]	No (abstract only)	Framework proposal / worked example	Law enforcement, security research [26]	Not specified [26]	Compliance-by-design (lifecycle) [26]
Habib Ahmed (2026) [27]	Yes	Qualitative framework proposal	Financial services [27]	All risk tiers [27]	Risk-based regulatory design [27]
A. Ranjbar et al. (2024) [28]	Yes	Standards review / organisational gap analysis	Healthcare (hospital) [28]	Not specified [28]	ISO 42001 gap analysis [28]
W. Finch & Marya Butt (2025) [13]	Yes	Hybrid systematic–scoping review	Cross-sector (healthcare, finance) [13]	All risk tiers / GPAI [13]	Compliance gap analysis / conceptual lens [13]
L. Judijanto (2025) [16]	Yes	Qualitative literature review	Healthcare, law enforcement, critical infrastructure, education [16]	High-risk [16]	Compliance challenge mapping [16]
Sung Une Lee et al. (2026) [29]	No (abstract only)	Framework proposal / case studies	General (executives, managers, developers) [29]	Not specified [29]	Risk assessment checklist / question bank [29]
S. Shojaei et al. (2026) [30]	Yes	Qualitative doctrinal legal analysis / case study	Healthcare (radiology) [30]	High-risk [30]	Post-market monitoring (legal mapping) [30]
Mazene Ameur et al. (2026) [31]	No (abstract only)	Framework proposal / validation	Telecommunications [31]	Not specified [31]	Automated compliance verification [31]
B. Jørgensen & Z. Ma (2025) [7]	Yes	PRISMA-ScR scoping review	Building energy management (BEMS) [7]	High-risk [7]	Compliance-by-design (design principles) [7]

Study	Full text retrieved?	Study type	Primary sector	Risk tier focus	Compliance orientation
W. Gehin et al. (2026) [9]	Yes	Institutional framework proposal / early implementation	Healthcare (cancer centre, France) [9]	High-risk [9]	Lifecycle governance / deployer compliance [9]
Mohammed Rashed & Y. Essa (2025) [32]	Yes	Product case study	Consumer IoT / smart home [32]	High-risk / GPAI [32]	After-the-fact documentation assessment [32]
Hasan Abu-Rasheed & L. Roddeck (2026) [33]	Yes	Interdisciplinary conceptual study / framework proposal	Education (automated grading) [33]	High-risk [33]	Compliance-by-design / lifecycle XAI [33]
G. M. Matilla et al. (2025) [34]	Yes	Software toolkit proposal / proof-of-concept	General (high-risk AI) [34]	High-risk [34]	After-the-fact bias detection [34]
Federica Picogna et al. (2026) [35]	Yes	Framework proposal / illustrative empirical case studies	Criminal justice, public education, banking [35]	High-risk (fairness auditing) [35]	Post-deployment fairness auditing [35]
A. Spagnolli et al. (2025) [36]	Yes	Empirical survey-based evaluation	Healthcare [36]	High-risk [36]	Transparency documentation (IFU) [36]
Sašo Karakatič et al. (2026) [5]	Yes	Quantitative factorial experiment / real-data validation	High-risk (employment, criminal justice, healthcare, finance) [5]	High-risk [5]	Fairness auditing methodology [5]
Rogers Kaliisa et al. (2025) [37]	Yes	Comparative policy analysis	Education / learning analytics [37]	Not applicable [37]	Regulatory comparison [37]
A. Barucci et al. (2026) [38]	No (abstract only)	Review / framework proposal	Clinical research [38]	High-risk [38]	Lifecycle alignment (MDR + AI Act) [38]
P. Radanliev (2025) [39]	Yes	Framework proposal / literature review	Healthcare / wearables [39]	Not specified [39]	Compliance-by-design (ethical governance) [39]
Radina Stoykova et al. (2024) [40]	No (abstract only)	Legal/technical challenge analysis	Law enforcement (Norwegian police) [40]	Not specified [40]	Techno-legal code of conduct [40]
P. Cañas et al. (2024) [41]	Yes	Framework proposal / documentary mapping	CCAM (automated mobility) [41]	High-risk [41]	Transparency documentation [41]
Á. Sánchez-García et al. (2025) [42]	No (abstract only)	System design proposal	Healthcare (CDSS) [42]	High-risk [42]	Post-deployment monitoring [42]

Study	Full text retrieved?	Study type	Primary sector	Risk tier focus	Compliance orientation
S. Reddy & MLOps (2026) [8]	Yes	Framework proposal / empirical benchmark evaluation	Finance, healthcare (future) [8]	High-risk [8]	Compliance-by-design (MLOps governance) [8]
F. Dennstädt et al. (2026) [11]	Yes	Regulatory analysis / framework proposal	Healthcare facilities [11]	All risk tiers (high-risk focus) [11]	Lifecycle compliance roadmap [11]
E. Bignami et al. (2025) [43]	Yes	Narrative review / commentary	Healthcare [43]	High-risk [43]	Compliance adaptation strategies [43]
Moritz von Zahn et al. (2025) [44]	Yes	Design Science Research / focus-group evaluation	Finance (fraud detection) [44]	High-risk (conditional) [44]	After-the-fact conformity assessment [44]
Avinash Agarwal & M. Nene (2025) [45]	Yes	Framework proposal / case-study validation	Cross-sector [45]	High-risk [45]	Five-layer governance framework [45]
Isabelle Hupont et al. (2023a) [14]	Yes	Comparative documentary assessment (Delphi)	Cross-sector (high-risk AI) [14]	High-risk [14]	Technical documentation assessment [14]
M. Karim (2026) [46]	Yes	Doctrinal / comparative legal analysis	Consumer collective redress / legal [46]	High-risk [46]	AI use-case legal mapping [46]
Jan Zacharias et al. (2022) [47]	Yes	Design Science Research / focus groups	Finance, healthcare (general ML) [47]	Not specified [47]	Explainable feature selection (pre-processing) [47]
Luca Nannini et al. (2024) [48]	Yes	Qualitative policy/document analysis	General EU / HR (illustrative) [48]	High-risk [48]	Explainability regulatory mapping [48]
E. Kanaki et al. (2026) [49]	No (abstract only)	Regulatory-technical analysis / scenario-based	Healthcare (6G) [49]	Not specified [49]	Regulatory compliance analysis [49]
Jesús Salgado-Criado (2025) [50]	Yes	Qualitative framework proposal / grounded theory interviews	Airlines, hotels, banking, retail [50]	All risk levels [50]	Human oversight (operational management) [50]
J. Kommusaar et al. (2025) [51]	Yes	Multi-method participatory roadmap development	Healthcare research (federated learning) [51]	Not specified [51]	Lifecycle roadmap (compliance-by-design) [51]
Ian Slesinger et al. (2024) [52]	Yes	Empirical qualitative co-creation study	Finance, biometrics, academia [52]	High-risk [52]	Participatory design for bias mitigation [52]

Study	Full text retrieved?	Study type	Primary sector	Risk tier focus	Compliance orientation
Alifia Hasan et al. (2025) [53]	Yes	Documentary mapping / standards review	Healthcare (US HDOs) [53]	Not specified [53]	Best-practice mapping [53]
Massimo Bottini et al. (2025) [54]	Yes	Comparative regulatory review	Healthcare / spine surgery [54]	High-risk [54]	Regulatory pathway guidance [54]
Jorge Francisco Aguirre-Sala (2025) [55]	Yes	Analytical review / framework proposal	Public administration [55]	High-risk [55]	Lifecycle governance (AIG model) [55]
A. Dragomir et al. (2026) [10]	Yes	Policy and Practice Review / doctrinal-comparative analysis	Public administration [10]	High-risk [10]	Human-centric governance framework [10]
Naga Srinivasulu Gaddapuri (2025) [56]	Yes	Empirical proof-of-concept evaluation	Financial services [56]	Not specified [56]	After-the-fact explainability-as-a-service [56]
Ronit Justo-Hanani (2026) [57]	No (abstract only)	Policy/regulatory analysis	Cross-sector [57]	Not specified [57]	Co-regulation analysis [57]
Devansh Gupta et al. (2026) [58]	No (abstract only)	Systematic literature review / framework proposal	FinTech / financial fraud detection [58]	Not specified [58]	XAI governance synthesis [58]
A. Ionică & Monica Leba (2026) [59]	No (abstract only)	Proof-of-concept empirical evaluation	Software audit / AI compliance detection [59]	Not specified [59]	Compliance-by-design (QINN architecture) [59]
Mircea Boșcoianu et al. (2026) [60]	No (abstract only)	Design Science Research / simulation	OT/ICS (energy, railway, manufacturing) [60]	Not specified [60]	Compliance-by-design / continuous assurance [60]
Sushanta Paul (2026) [61]	No (abstract only)	Documentary review / framework proposal	Telecommunications (5G/6G) [61]	Not specified [61]	Telecom-tailored lifecycle checklist [61]
K. Rajasekaran (2024) [62]	Yes	Mixed-methods (survey + expert interviews)	Enterprise software engineering [62]	Not specified [62]	Enterprise AI developer tools governance [62]
Tahmina Zebin et al. (2026) [63]	No (abstract only)	Evidence synthesis / framework mapping + worksheets	Healthcare [63]	Not specified [63]	Healthcare AI risk framework worksheets [63]
Sebastian Biewer et al. (2023) [64]	Yes	Formal method / proof-of-concept	High-risk AI (admissions, HR, credit, justice) [64]	High-risk [64]	Runtime fairness monitoring [64]

Study	Full text retrieved?	Study type	Primary sector	Risk tier focus	Compliance orientation
Janardhana Naidu Kola (2026) [65]	Yes	Framework proposal / empirical benchmark evaluation	Enterprise risk management / finance [65]	High-risk [65]	Compliance-by-design (conformal prediction) [65]
Elena Sblendorio et al. (2026) [66]	No (abstract only)	Empirical evaluation / framework feasibility test	Healthcare / nursing (LLM evaluation) [66]	High-risk [66]	Regulatory-aligned model evaluation [66]
Min Xu (2026) [67]	Yes	Empirical comparative multi-case study	Finance, technology, healthcare (multinationals) [67]	Not specified [67]	Strategic adaptation to regulatory fragmentation [67]

The body of literature is heavily skewed toward high-risk AI systems [1–3, 5, 6, 9, 11, 12, 23, 33], reflecting the Act's most stringent tier as the primary compliance challenge in the literature. Healthcare is the most commonly studied sector [9, 11, 25, 28, 38, 42, 43, 63, 66], followed by finance [3, 27, 44, 56], public administration [10, 55], and education [12, 33]. The majority of studies are framework proposals or proof-of-concept demonstrations rather than randomised or controlled trials, and very few include real-world organisational deployments with robust comparative designs. Of the 67 sources, 16 were available only as abstracts, which limits the depth of extraction possible for those studies.

Thematic Analysis

The compliance approaches identified across the 67 sources can be organised into six broad themes: (1) AI-assisted and automated compliance tools; (2) technical documentation and transparency frameworks; (3) fairness, bias detection, and explainability methods; (4) organisational governance and human oversight frameworks; (5) post-market and deployment monitoring; and (6) standards, harmonisation, and cross-regulatory integration.

Theme 1: AI-Assisted and Automated Compliance Tools

A distinct cluster of studies evaluates or proposes tools that use AI itself—particularly large language models and retrieval-augmented generation—to assist with compliance verification and documentation.

Sovrano et al. (2025) is among the few studies to empirically compare tools. Their five-phase documentation workflow evaluated DoXpert (a retrieval-augmented tool) and standalone ChatGPT against a 65-item checklist derived from AI Act Annex IV [1]. DoXpert achieved 81.54–83.08% accuracy and 83.33–84.06% F1 in iterative draft review, substantially outperforming standalone ChatGPT 4 (69.23–70.77% accuracy; F1 61.54–64.15%) [1]. Against expert-majority judgments, DoXpert reached 84.6–96.9% accuracy and 91.2–98.4% F1, compared with ChatGPT 4's 47.7–53.8% accuracy [1]. The authors caution that ChatGPT 4 exhibited more hallucinations, while DoXpert could fail due to keyword mismatch in retrieval [1]. Importantly, expert disagreement itself was substantial—21.53–32.30% across the two case studies—underlining that there is no unambiguous legal ground truth against which tools can be fully validated [1].

Cappelli et al. (2026) propose and evaluate a GPT-4-based EU AI Act Compliance Checker, tested across four high-risk Annex III cases. Automated evaluation showed strongest semantic alignment for regulatory obligations (BERT similarity 0.911; TF-IDF 0.747) but markedly weaker performance for legal risks (TF-IDF 0.186) and technical documentation (TF-IDF 0.199) [2]. Expert reviewers gave an average score of 3.93/5, with noted weaknesses in actor-role precision and audit-level legal detail [2]. The authors explicitly position the tool as a pre-audit aid, not a compliance certification instrument [2].

EUROCOMPLY (Ameur et al., 2026) extends the LLM-based approach to telecommunications, combining vector-based and graph-based retrieval to inspect datasets and AI/ML pipelines for alignment with the EU AI Act, GDPR, and 3GPP standards [31]. Validation on 20 telecom use cases is described as showing high faithfulness and strong performance via expert assessments and LLM-as-a-Judge evaluation, though no specific metrics are reported in the available abstract [31].

The EaaS platform proposed by Gaddapuri (2025) applies a different LLM-integration strategy: model-agnostic post-hoc explainers (LIME, SHAP, Anchors) interface with black-box financial models via APIs, and their outputs are projected onto a regulatory knowledge graph covering EU AI Act obligations [56]. The reported regulatory coverage reached 100% for loan approval and credit scoring but fell to 92% for fraud detection [56]. Compliance approval time was reduced by 57.1–59.7% across the three use cases [56], and 86% of surveyed compliance officers preferred the system-generated explanations to manual alternatives [56]. These results are promising but derive from three isolated model evaluations without a control group.

Ionică & Leba (2026) take a structurally different approach, embedding quality-function-deployment (QFD) knowledge into neural-network topology to create Quality-Informed Neural Networks (QINN), demonstrating cross-validated AUC of 99.47% on an AI-enabled code-detection task [59]. The claim that the resulting explanations align with EU AI Act documentation obligations is conceptual rather than validated through regulatory assessment [59].

The Boşcoianu et al. (2026) Unified Compliance Framework (UCF) maps 47 atomic regulatory requirements, including the AI Act, to standards and auditable evidence artefacts through a Continuous Assurance Loop, demonstrated through digital-twin simulation with a simulated ACRI improvement from 45–61 to 65–73 across OT threat scenarios [60]. This remains simulation-only evidence with no field validation [60].

Across this theme, a clear pattern emerges: LLM-assisted tools perform best on structured, high-coverage tasks such as identifying whether documentation addresses known regulatory obligations, but struggle with nuanced legal reasoning, cross-regulatory scope, and technical conformity verification [1, 2]. None of the tools has been independently validated in a real production compliance context, and all authors recommend retaining human legal expertise in the compliance loop [1, 2].

Theme 2: Technical Documentation and Transparency Frameworks

A large number of studies address the gap between abstract legal documentation obligations and practical, operationalisable documentation artefacts.

Hupont et al. (2023a) assessed six existing documentation approaches—including Datasheets for Datasets, Model Cards, and AI Factsheets—against EU AI Act technical information requirements, using a five-expert JRC Delphi procedure [14]. Datasheets for Datasets was judged most complete for dataset documentation, while AI Factsheets was most complete for AI-system documentation; however, none fully provided the technical depth required by conformity assessment bodies [14]. The authors recommend combining and extending existing approaches with greater technical depth, automation tools, and tailored guidance for high-risk system contexts [14].

Hupont et al. (2023) also proposed Use Case Cards—a UML-based template paired with a visual diagram—as a preliminary tool for documenting intended use and assessing risk tier [24]. Expert questionnaire evaluation (n=11) yielded a mean AI Act contextualisation rating of 4.18/5 and risk-assessment utility of 3.82/5 [24], though the authors are explicit that the cards do not constitute exhaustive compliance documentation [24].

Cañas et al. (2024) adapted the Model Card concept for cooperative, connected, and automated mobility (CCAM), mapping 21 proposed Model Card information items to AI Act Articles 9–15 and JRC/ALTAI trustworthiness criteria, and proposing a companion MLOps Card for infrastructure and lifecycle documentation [41]. The contribution is documentary and normative rather than empirically validated [41].

Spagnolli et al. (2025) took an empirical approach to Article 13 transparency, administering an online survey to 813 healthcare stakeholders to test whether users' transparency needs map onto the mandatory 12-section Instructions for Use (IFU) structure [36]. Sixteen of 24 assessed needs had a preferred IFU section selected by fewer than half of participants [36], and some needs—such as handling medical-case complexity and preserving human contact—did

not appear to map to any section [36]. The authors recommend supplementing the mandatory IFU with a navigation interface, stakeholder-specific signposting, and locally completed context-specific information [36].

Manganello et al. (2025), applying the XAI-ED Consequential Assessment Framework to 14 Italian universities' governance documents, found that technical robustness and safety was the lowest-scoring dimension (mean 19.64/100), while societal well-being was highest (52.38/100) [12]. Institutional Index scores ranged from 0.00 to 60.32, indicating wide heterogeneity among early adopters [12]. Importantly, only 13% of Italian institutions had adopted AI policies by September 2025 [12], and high governance scores on documentation did not necessarily indicate actual implementation [12].

Across this theme, documentation approaches converge on the need to move beyond aspirational statements toward operational mechanisms, particularly for technically demanding requirements such as robustness testing, bias monitoring, and validity evaluation [12, 14, 33]. The gap between user-accessible documentation and the technical detail required for conformity assessment bodies is a persistent finding [14, 14].

Theme 3: Fairness, Bias Detection, and Explainability Methods

A substantial portion of the included studies addresses technical methods for demonstrating fairness and explainability, which are central to the Act's high-risk requirements under Articles 9–15.

Mahajan et al. (2025) provide the most detailed quantitative evaluation of bias mitigation under AI Act constraints, using 12.4 million insurance observations from four European carriers [3]. Comparing re-weighing, reject-option classification, and adversarial debiasing, only adversarial debiasing closed 65–82% of premium gaps while keeping Max-EOG below the 0.10 materiality threshold, adding only 14 basis points to SCR at the optimal training weight λ=0.40 [3]. Re-weighing reduced maximum unfairness by 44% but sacrificed approximately 15% of the accuracy uplift and imposed a larger capital impact [3]. The authors conclude that proactive adversarial debiasing is capital-efficient and becomes economically preferable once detection probability exceeds 8.9% [3].

Karakatič et al. (2026) identify a fundamental methodological hazard in fairness auditing: when the same biased labels used for training are also used for evaluation, accuracy can be overstated by 4.50% on average, F1 understated by 10.10%, and AUC understated by 13.74% [5]. Most strikingly, SMOTE appeared to reduce accuracy by 3.44% under biased evaluation but improved it by 7.96% against clean labels—a reversal of 11.39 percentage points [5]. The authors recommend that regulators require evaluation-bias detection and uncertainty quantification before fairness metrics are accepted as compliance evidence [5].

Picogna et al. (2026) address the problem of fairness-measure selection, proposing a four-question decision workflow implemented in JASP for Audit, demonstrated through COMPAS, DUO, and German Credit Risk case studies [35]. In the DUO case, switching from one measure to another changed the conclusion from non-significant to significant discrimination, illustrating that measure choice materially affects compliance conclusions [35]. The authors note that their workflow is limited to binary classification and cannot address individual or causal fairness [35].

Matilla et al. (2025) propose the humancompatible.detect toolkit, which implements Maximum Subgroup Discrepancy (MSD) for intersectional bias detection and subsampled $\ell\infty$ for threshold testing, claiming linear sample complexity in protected attributes and provably optimal subgroup identification [34]. The demonstration is on a synthetic example, and no real-world compliance evaluation is reported [34].

For explainability, Kalra (2025) reports that deploying SHAP and LIME in CCaaS contact centre systems, combined with continuous bias monitoring and cryptographically hashed audit trails, was associated with a 67% improvement in compliance audit readiness and an 81% reduction in regulatory findings across three multinational organisations [23]. However, no control group was used, and the evaluation period of 12–18 months may not capture long-term sustainability [23].

Zacharias et al. (2022) propose and evaluate SHAP-based explainable feature selection through two focus groups (n=12), finding overall effectiveness rated 3.93/5 and stakeholder-justification ability 4.01/5 [47]. SHAP-based selection was not clearly superior to benchmark methods on predictive performance but was assessed as ”at least not worse” [47].

Von Zahn et al. (2025) develop a semi-automated conformity-assessment framework evaluated through five focus groups (n=21) in a financial fraud-detection scenario, incorporating statistical parity difference, equalized odds dif-

ference, explanation stability, and data drift [44]. Focus-group respondents strongly agreed the design requirements were necessary [44]. The illustrative fraud-detection report rated fairness green, explainability red, and performance yellow [44] —demonstrating the tool's diagnostic function but also highlighting that simultaneous compliance on all dimensions is not guaranteed.

The Reddy & MLOps (2026) framework enforces a governance approval gate requiring accuracy, fairness (demographic parity difference and equalized odds difference each ≤0.05), and full SHAP coverage [8]. On the German Credit Dataset (1,000 records), MLP was the only model to receive unconditional approval (DPD 0.04 ± 0.01; EOD 0.05 ± 0.01), while LR and RF were rejected and XGBoost received conditional approval pending bias-aware resampling [8]. The one-dataset proof-of-concept limits generalisability [8].

Across this theme, a recurring tension emerges between fairness, accuracy, and capital efficiency. Mathematical incompatibility between fairness criteria—most clearly articulated by Kola (2026), who notes that closing statistical parity gaps worsens equalized-odds disparity [65] —means that selecting a fairness criterion is itself a governance decision, not merely a technical one. The finding from Karakatič et al. (2026) that evaluation data contamination can reverse the apparent effectiveness of interventions such as SMOTE is particularly important for compliance practice, as it suggests that currently accepted fairness-auditing workflows may produce systematically misleading results [5].

Theme 4: Organisational Governance and Human Oversight Frameworks

The second-largest cluster of studies addresses organisational-level governance structures, human oversight obligations under Article 14, and institutional readiness.

The Kauffman et al. (2026) pilot with 11 manufacturing companies in the UK and Brazil is among the few studies to report pre/post effectiveness with statistical testing. Their compliance preparedness framework—combining risk-classification decision trees, obligation checklists, scenario-based workshops, documentation templates, and assigned compliance roles—was associated with a rise in risk-identification accuracy from 40% to 86%, regulatory knowledge from 42% to 81%, and self-reported confidence from approximately 2.58 to 4.09/5 (all $p<0.01$, Cohen's d 1.0–1.5) [4]. The authors caution that the 11-company, four-week pilot without a control group limits causal attribution [4].

Enang et al. (2026) introduce the Authentic Intelligence framework, mapping Article 14 human oversight requirements to visibility/intervention, contextual interpretation, override/escalation, and feedback/adaptation [22]. Drawing on 50 senior-manager interviews and 685 JRC AI implementation cases, the framework identifies 12 empirically derived failure modes, with six showing high correspondence (≥20% JRC citation rate) and six moderate correspondence [22]. The authors recommend risk-calibrated governance: routine decisions can use lighter controls, consequential decisions require critical human intervention, and contested decisions require critical requirements across all four mechanisms [22].

Lee et al. (2026) develop an expert-informed HCAI pre-design governance package combining AHP surveys (n=28) with think-aloud interviews (n=15) [21]. Human oversight and control received the highest global priority among governance criteria, followed by personal information protection, design ethics, intellectual property rights protection, and limits of algorithmic autonomy [21]. Experts framed trustworthy AI governance as a problem of controllability, responsibility allocation, and verifiable human intervention rather than model performance [21], though the package is explicitly not a validated legal compliance model [21].

Gehin et al. (2026) describe operationalising EU AI Act deployer obligations at a French comprehensive cancer centre through a dual-axis risk stratification model, a five-category AI-related adverse-event taxonomy, and structured proficiency testing against simulated AI errors [9]. Early implementation involved review of approximately ten projects over six months, but no formal AI-adverse-event reports or proficiency-testing data had yet been generated [9]. The authors conclude that institutional readiness depends more on governance capacity than algorithmic performance [9].

Dragomir et al. (2026) develop a six-dimensional governance framework for AI in public administration—legal anchoring, accountable discretion, fundamental rights by design, meaningful human oversight, contestability and justification, and institutional resilience—through legal-doctrinal analysis and comparative documentary study of the Netherlands, France, and Estonia [10]. Their central finding is that formal compliance does not by itself ensure

lawful administration; accountability must follow the chain of influence through which algorithmic systems shape outcomes, not only the formal chain of decision-making [10].

Salgado-Criado (2025), drawing on ten executive interviews across service industries, argues for treating oversight as organisational "meta-control," distinct from participation in the AI decision process itself [50]. The paper identifies five design dimensions—decision significance, oversight goals, system complexity, overseer skills, and organisational culture—and finds that banking-sector managers were the only respondents responsible for formally high-risk AI Act applications [50]. The study notes that oversight design must be application-specific and cannot be addressed by a single prescribed framework [50].

Aguirre-Sala (2025) proposes a lifecycle-based AI governance model for public administration intended to complement the EU AI Act, ISO/IEC 42001, and NIST AI RMF, but concludes that existing frameworks leave gaps in public assessment, affected-person challenges, and external oversight [55]. The paper calls for binding AI Impact Assessment procedures and recommends against relying on ISO 42001 alone because it excludes user and citizen participation [55].

Dennstädt et al. (2026), in a healthcare-specific synthesis, propose a 10-step, four-phase compliance roadmap for deployers, and find that only 26% of surveyed hospital representatives felt ready for implementation [11]. The roadmap identifies mandatory obligations—including Fundamental Rights Impact Assessments, local pre-go-live validation, and structured human oversight mechanisms—and distinguishes them from best practices such as governance committees and vendor due-diligence audits [11].

Across this theme, the convergence point is that technical tools alone are insufficient: compliance requires institutional capacity, assigned accountability, trained oversight personnel, and governance structures that embed compliance across the AI lifecycle [9–11]. The persistent weakness of aspirational documentation relative to operational mechanism implementation—evidenced both quantitatively in the Italian university study [12] and qualitatively across multiple sectors—suggests that organisations often satisfy the appearance of compliance more readily than its substance.

Theme 5: Post-Market and Deployment Monitoring

Several studies specifically address the post-deployment monitoring obligations of the AI Act (Articles 72–73, 26(5)) and how to implement them in practice.

The ESR consensus recommendations for AI medical device post-market surveillance (Cuocolo et al., 2025) represent an expert-validated set of practices developed through a two-round Delphi procedure, with 12 of 13 items reaching the 75% consensus threshold [25]. Key recommendations include deploying a PMS platform available at the point of system launch and updated with device changes, conducting periodic provider reviews every 6–12 months, displaying baseline accuracy and uncertainty metrics, and using interoperable shared platforms for monitoring multiple devices [25]. The panel emphasised that relying on voluntary individual reporting is insufficiently reliable, and recommended institutional-level data collection [25].

Shojaei et al. (2026), through a doctrinal legal analysis of a prostate-cancer radiology AI SaMD, map ten post-market obligation categories across the EU AI Act and MDR, finding convergence in documentation and performance monitoring but divergence in human oversight (primarily AI Act) and trend reporting (primarily MDR) [30]. The authors identify workforce and workflow demands—including digital literacy requirements, possible dedicated personnel, and the sustainability of clinician vigilance—as practical barriers to meaningful human oversight [30].

Sánchez-García et al. (2025) propose ShinAI-Agent, a modular OLAP-architecture monitoring system for health AI that disaggregates confusion-matrix-derived performance and fairness metrics across time and sensitive subgroups, combined with dataset-shift monitoring and model-retraining advisory [42]. The abstract does not report validation results [42], but the system directly addresses the non-additivity of confusion-matrix-derived metrics across subgroups—a technical barrier identified in the literature [42].

Kola (2026) applies conformal prediction theory to enterprise AI governance, proposing confidence-aware escalation tiers that jointly evaluate prediction-set cardinality and risk category to route outputs to automated action (Tier 1), human review (Tier 2), or mandatory human decision (Tier 3) [65]. On the FICO HELOC benchmark, 82.8% of

decisions fell to Tier 1 and 17.2% to Tier 2 at α=0.05, with empirical coverage of 0.9474 against a theoretical 0.9500 [65]. The framework maps mechanisms explicitly to AI Act Articles 9, 12, 14, and 15 [65].

Across this theme, a key tension is between scalability and meaningfulness of oversight: Shojaei et al. (2026) note that reviewing every AI-triaged case would undermine the efficiency rationale for semiautonomous systems, while Kola's (2026) conformal approach offers a principled way to calibrate automation rates to verifiable error-rate bounds [65]. Monitoring obligations for drift, bias, and performance are widely discussed but rarely operationalised with demonstrated reliability in real deployments [42, 65].

Theme 6: Standards, Harmonisation, and Cross-Regulatory Integration

Several studies examine the role of harmonised technical standards and multi-framework integration in enabling practical compliance.

Kilian et al. (2025), through 23 qualitative interviews with EU AI organisations and analysis of CEN-CENELEC JTC 21 draft standards, provide the most detailed treatment of standards-based compliance [15]. Only 8 of approximately 35 planned work items had been published as of the study period, with interviewees estimating that meeting even a single standard typically requires 12 months [15]. Annual compliance personnel costs were estimated at approximately €100,000, with certification costs exceeding €200,000 in some medical-tech firms [15]. The paper identifies participation imbalance in standardisation committees—where large enterprises and major technology companies predominate—as a structural barrier to SME influence [15].

Finch & Butt (2025), in the broadest systematic review in this collection, develop the conceptual lens of "compliance asymmetry" to describe how structural frictions between regulatory ambition and institutional capacity produce unequal compliance outcomes [13]. They note that ALTAI reportedly overlooks 69% of known AI security vulnerabilities [13], ISO/IEC 42001 imposes documentation and auditing burdens that can exceed SME capacity, and NIST AI RMF has low SME uptake [13]. The review maps actor-specific obligations across the value chain, noting that downstream importers, distributors, deployers, and public actors can face verification duties without access to system architecture or training data [13].

Ranjbar et al. (2024) apply ISO/IEC 42001 through a hospital gap analysis, identifying weaknesses in AI lifecycle management capability, data governance, preparation, and workforce capacity [28]. The authors note that the standard is proposed as a future harmonised standard for the Act but that healthcare organisations may have insufficient technical debt in modern software engineering and data-management practices to implement it effectively [28].

Jørgensen & Ma (2025) identify, in their BEMS scoping review, that EU AI Act compliance for building energy systems must integrate with GDPR, NIS2, the Cybersecurity Act, and EPBD—four overlapping regulatory layers with different objectives and mechanisms [7]. The paper recommends treating compliance as compliance-by-design from the outset, rather than retrofitting controls [7].

Ahmed (2026) and Habib Ahmed (2026) both extend the EU AI Act framework to financial services, mapping risk categories and corresponding regulatory responses across unacceptable, high, limited, and minimal risk, and noting that the financial sector is not explicitly listed in Annex III but that some Annex III risks may apply [27]. The proposed integration of AI Act governance with financial-services law is a conceptual framework without empirical validation [27].

Justo-Hanani (2026) examines the co-regulatory model underlying the Act, arguing that effective regulation requires mandated data-sharing mechanisms between private and public actors to build regulatory capacity, and that the framework must be flexible enough to avoid industry capture [57].

Synthesis

The Central Heterogeneity: Effectiveness Evidence Across Method Types

The most important observation across these 67 sources is that the question of which methods and tools are "most effective" cannot be answered uniformly, because effectiveness is defined differently, measured differently, and

achieved under different organisational and technical conditions across studies. Heterogeneity in findings is substantial but explicable through three systematic patterns: the compliance lifecycle stage at which a method operates, the organisational capacity required to deploy it, and the specificity of the regulatory obligation being addressed.

Lifecycle Stage Distinctions

Methods operating at the pre-design and design stage—including the HCAI pre-design governance package [21], compliance-by-design MLOps frameworks [6, 8], and adversarial debiasing during model training [3] —address a fundamentally different compliance problem than post-deployment tools such as runtime fairness monitors [64], post-market surveillance platforms [25, 42], or LLM-based documentation checkers [1, 2]. Studies operating at earlier lifecycle stages consistently report that compliance is less costly and more technically tractable when controls are embedded during development rather than added retrospectively [6–8, 33]. This finding is methodologically coherent: compliance-by-design prevents the technical debt described by Ranjbar et al. (2024) [28] and the legacy-infrastructure integration barriers identified by Abu-Rasheed & Roddeck (2026) [33] from accumulating. However, the evidence for pre-design approaches is predominantly conceptual—none of the compliance-by-design frameworks in this collection has been field-validated against actual regulatory outcomes.

Conversely, post-deployment monitoring studies (the PMS consensus by Cuocolo et al., 2025 [25]; Shojaei et al., 2026 [30]; Sánchez-García et al., 2025 [42]) necessarily address a different compliance phase, and direct comparison of their effectiveness with pre-design methods is not meaningful.

Study Quality and Methodological Hierarchy

When weighting the available evidence by methodological rigour, the findings are as follows. Only Kauffman et al. (2026) and Mahajan et al. (2025) provide pre/post or comparative quantitative evidence with statistical testing in real-world contexts. Kauffman et al.'s pre/post pilot (n=11 companies) shows large and statistically significant improvements ($p<0.01$, Cohen's d 1.0–1.5) [4] from a structured compliance preparedness framework, but is limited by self-selection, the absence of a control group, and a four-week evaluation window [4]. Mahajan et al.'s insurance data study (n=12.4 million observations) provides the most rigorous quantitative evidence for any specific technical method, finding that adversarial debiasing dominates re-weighing and reject-option classification on the fairness-accuracy-capital tradeoff in this domain [3]. However, this finding is specific to life and health insurance underwriting, and the debiasing is single-shot, requiring continuous monitoring for ongoing compliance [3].

The large majority of remaining studies are framework proposals or proof-of-concept evaluations in controlled or synthetic settings, with no control group, no real regulatory deployment, and often very small evaluation samples. The LLM-based tools (Sovrano et al., 2025; Cappelli et al., 2026 [1, 2]) are evaluated against expert judgments that themselves show substantial disagreement [1], and against documentation derived from open-source AI systems rather than real organisational compliance workflows [1]. The practitioner focus-group evaluations of von Zahn et al. (2025) [44] and Zacharias et al. (2022) [47] are encouraging but involve controlled, artificial settings with small samples geographically concentrated in Germany and Switzerland [44]. Accordingly, claims about effectiveness for any specific tool or method should be treated with caution proportionate to the methodological rigour of the underlying study.

Sector and Population Context

Several apparently conflicting findings across studies resolve when contextual differences are accounted for. The Kauffman et al. (2026) pilot in non-EU manufacturing companies achieved large improvements in compliance readiness [4], while Judijanto (2025) and Finch & Butt (2025) document persistent compliance failures across EU member states [13, 16]. These findings are not contradictory: Kauffman et al.'s participants were starting from a very low baseline (40% risk-identification accuracy before intervention) [4] in an intentionally supportive four-week framework environment, while Judijanto and Finch & Butt describe the broader population of organisations attempting compliance without structured support. Similarly, the high audit-readiness improvements reported by Kalra (2025) in CCaaS multinational enterprises [23] are specific to large organisations with 12–18 AI models and resources to deploy 7.3 FTE and spend 6.2 months on implementation [23] —a very different context from SMEs, who face disproportionate barriers across multiple studies [13, 15, 16, 43].

Confounder: The Documentation-Implementation Gap

A systematic confounder across the governance and documentation literature is the gap between documented aspirations and operational implementation. Manganello et al. (2025) quantify this directly: technical robustness and safety scored a mean of 19.64/100 in governance documents [12] despite universities often claiming AI ethics commitments. The Finch & Butt (2025) review identifies ALTAI as potentially enabling "ethics-washing" because its voluntary, non-binding character allows organisations to present documented commitments without implementing technical safeguards [13]. Dragomir et al. (2026) make the same point for public administration: formal compliance with documentation requirements does not by itself ensure that authorities can understand, justify, correct, suspend, or democratically control the systems they deploy [10]. This confounder suggests that studies measuring compliance by documentation scores or self-reported readiness may overestimate real-world compliance capacity.

What Can Be Concluded for Specific Contexts

For organisations developing high-risk AI systems and seeking technical compliance with Articles 9–15, the evidence most strongly supports three practices. First, embedding fairness testing and explainability into training pipelines (adversarial debiasing, SHAP-based monitoring, governance gates) is preferable to retrofitting these capabilities post-deployment, particularly in regulated sectors such as insurance [3, 8]. Second, fairness audits should include explicit evaluation-bias detection and uncertainty quantification before fairness metrics are treated as compliance evidence, given the systematic distortions documented by Karakatič et al. (2026) [5]. Third, LLM-based documentation tools can usefully identify gaps in technical documentation but should not be used as standalone compliance checkers, and always require legal expert validation [1, 2].

For deployers—particularly healthcare facilities, public authorities, and educational institutions—the evidence most clearly supports structured institutional governance: designated oversight roles, multidisciplinary steering committees, risk-stratified system inventories, documented human override authority, and continuous monitoring protocols [9–11]. The clinical AI literature emphasises that deployers often underestimate their own obligations and lack the regulatory familiarity to discharge them [25, 30].

For SMEs and organisations in non-EU jurisdictions exporting AI-embedded products to the EU, the Kauffman et al. (2026) scenario-based training and structured framework pilot offers the most directly applicable evidence of achievable readiness improvement within a short timeframe, albeit from a single study with the design limitations noted above [4]. The Kilian et al. (2025) findings on standardisation burdens indicate that SMEs face structurally greater costs and participation barriers than larger organisations in navigating harmonised technical standards [15].

For sectors with sector-specific dual regulation—healthcare (AI Act plus MDR), financial services (AI Act plus sectoral financial law), and dual-use defence systems (AI Act plus NATO frameworks)—the evidence consistently shows that compliance approaches must be integrated across regulatory frameworks rather than treating AI Act obligations in isolation [17, 27, 30]. The Shojaei et al. (2026) mapping of AIA and MDR post-market obligations illustrates that treating these frameworks separately creates ambiguity in human oversight responsibilities and incident reporting [30].

The single most important capability gap identified across the included literature is the absence of field-validated, sector-specific compliance tools that integrate the full AI Act lifecycle—from risk classification and technical documentation through conformity assessment and post-market monitoring—with adequate performance across the SME capacity range. Every tool and framework evaluated in this collection either addresses only a subset of the compliance lifecycle, has been validated only in controlled or synthetic settings, or requires organisational resources that are unavailable to low-capacity actors. This gap is structural: it reflects the compliance asymmetry between regulatory ambition and institutional capacity that Finch & Butt (2025) identify as the central unresolved design problem in EU AI Act implementation [13].

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